

Exercise 4 (NHANES)

We keep using the NHANES data from your R Data Project.

Task 1 ()

1. How strong is the relationship between BMI and systolic blood pressure? Is it significant? How much of the variation in systolic blood pressure can be explained (in a statistical sense) by variation in BMI?

For the following exercise, create a binary age (younger: $age \leq 50$ and older: $age > 50$) and a binary smoking variable.

2. Does the relationship between systolic blood pressure and BMI change when you adjust for age (categorized)? Interpret the coefficients of the resulting model (when you mean-center BMI before fitting the model, you can also interpret the intercept). Would you say that BMI has a clinically relevant impact on blood pressure, according to your model?
3. Try to find a better model to predict systolic blood pressure by including more covariates. Select a number of candidate covariates which in your opinion may be related to systolic blood pressure, and then choose a model selection strategy and a criterion/test for comparing models. Describe the model with the best fit according to your search, and interpret the model coefficients.